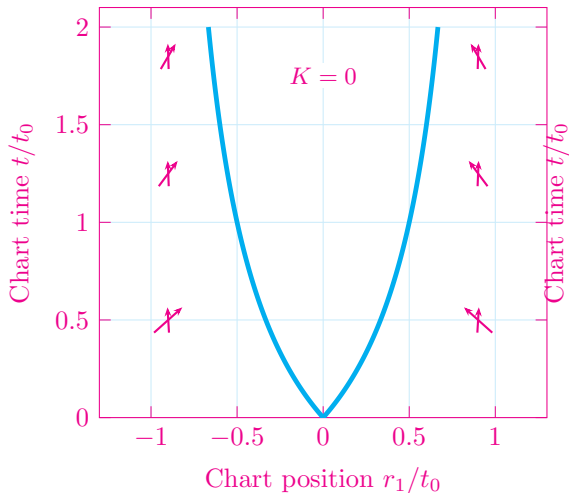


Flat: exact coordinate change

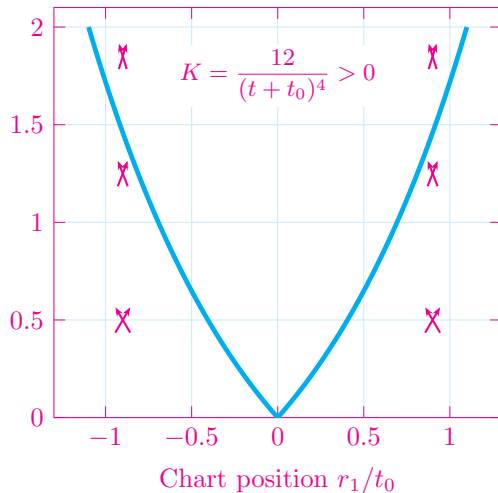


$$ds^2 = dt^2 - d(\lambda \mathbf{r})^2$$

$$\frac{dr_1}{dt} = \frac{\pm 1 - r_1/t_0}{\lambda}$$

$\lambda(t) = 1 + t/t_0$, $t_0 > 0$: the same chart and the same scale factor.

Curved: assigned displacement



$$ds^2 = dt^2 - \lambda^2 d\mathbf{r}^2$$

$$\frac{dr_1}{dt} = \frac{\pm 1}{\lambda}$$

Thick curves: light rays from the origin. Short arrows: local light directions.
Curved coordinate paths do not prove curvature; the four-dimensional K does.